

DOKUZ EYLÜL UNIVERSITY
ENGINEERING FACULTY
DEPARTMENT OF COMPUTER ENGINEERING

**A Mobile Application Providing Personalized
Interaction with Smart Home IoT Devices**

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October, 2025
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A Mobile Application Providing Personalized Interaction with Smart Home IoT Devices

**A Thesis Submitted to the
Dokuz Eylül University, Department of Computer Engineering
In Partial Fulfillment of the Requirements for the Degree of B.Sc.**

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CHAPTER ONE

INTRODUCTION

1.1 Background Information

This study focuses on the interaction between Smart Home systems and Internet of Things (IoT) technologies, which are central to modern residential technologies. Specifically, the design and development processes of personalized mobile applications that enhance user interaction with these systems have been examined.

1.1.1 Conceptual Development and Historical Origins

The emergence of the smart home concept has a history older than today's integrated systems and is fundamentally based on the need for comfort and automation. The first attempts at home automation were carried out in the 1960s.

Early Automation Initiatives: The first prominent example from this period is considered to be the ECHO IV (1966) system, which had basic functions such as in-home temperature control and grocery list creation, but was not commercially produced (Lutolf, n.d.). Through these systems, the goal was to simplify tasks within the residence. The X10 protocol, which enabled communication between home devices over the electrical wiring, was developed in 1969 and made significant contributions to the widespread adoption of automation.

Precedence of M2M Technology: The fundamental technology that played a critical role in the smart home systems' transition to the modern era is the M2M (Machine-to-Machine) communication concept. M2M is defined as an infrastructure that allows devices to exchange data with a central server or other devices without human intervention (Jiang, 2012). This technology, which can be described as inter-device communication, is considered the first implementer of the core principle of IoT and has been successfully used in the fields of smart meters and industrial automation.

1.1.2 Internet of Things (IoT) and Smart Home Concepts

The Internet of Things (IoT), which evolved from M2M and spans a much wider scope, is defined as a global network architecture that enables physical objects (sensors, devices, and software) to connect, collect, and share data via the internet (Ahmad, 2014).

Smart Home Definition: Residences where in-home systems (security, heating, lighting, etc.) are automated and remotely manageable through the use of IoT devices are called Smart Homes. Devices in smart homes are designed to be responsive to environmental context (user presence, weather conditions), thereby enabling automated decision-making.

The Role of Mobile Applications: In IoT-based smart home systems, mobile applications serve as a critical interface between the user and the complex device ecosystem. This has made it possible for users to monitor and control their homes from anywhere in the world via their smartphones.

1.2 Problem Definition

1.2.1 Identification and Significance of the Problem

The main objective of Smart Home systems is to improve the quality of residential life and ensure operational efficiency. Although inter-device connectivity and automation have been achieved through Internet of Things (IoT) technologies, significant shortcomings have been identified in the quality of user interaction with existing systems. The core problem to be addressed is the inadequacy of smart home systems in proactive personalization and contextual feedback capabilities.

Devices in current systems generally operate in two ways:

Pre-programmed Scenarios: Devices adhere to predefined rules for specific times or simple triggers (e.g., sunset).

Direct User Commands: Devices execute instantaneous commands (e.g., "turn on the light") transmitted via mobile applications.

This study is based on the assumption that these types of reactive and static operating modes fail to meet the expectations of modern smart home users. Smart homes need to be elevated to a level where they can anticipate needs that users themselves may not even be aware of.

1.2.2 Inadequacy of Current Solutions

Although currently available mobile applications and system interfaces on the market provide smart home users with extensive control capabilities, critical points where these solutions fall short have been identified:

Lack of Personalization: Existing systems treat every user as a uniform entity. The fact that different users in the home (e.g., parents and children) have different habits at different times has been overlooked. Consequently, the scenarios offered by the system cannot go beyond a singular user or an instantaneous state.

Inability to Predict Habits: Despite the flow of data between devices, a mechanism for processing this data with complex algorithms to enable habit prediction has not been established. For instance, contextual information such as what a user's lighting preference will be before dinner on a weekday or how the thermostat setting will change on weekend mornings cannot be learned by the system.

Inability to Generate Feedback and Recommendations: Systems have typically only executed the command and relayed status information (e.g., "the lamp is on") to the user. However, they have been unable to offer proactive feedback or energy-saving recommendations based on the current state of the devices or the user's past behavior. For example, informative and actionable output such as, "Your air conditioning usage at this hour is 30% higher than normal; the recommended setting is as follows," cannot be provided.

These shortcomings make the interaction of smart home users with their systems unengaging and limited, preventing the full potential of comfort and efficiency offered by the system from being reached. This thesis aims to find a solution to this problem area by integrating Machine Learning (ML)-based prediction algorithms.

1.3 Motivation

1.3.1 Rationale for Solving the Problem and Societal Contribution

The primary motivation for this study is the potential to elevate Smart Home systems from being a reactive control mechanism to the level of a proactive life assistant. Addressing the gaps in the personalization and prediction capabilities of existing systems will directly translate into improved user experience and societal benefit.

The greatest advantage gained by solving this problem is enabling residents to lead more efficient and optimized lives without even realizing it.

Quality of Life Optimization: By predicting daily habits and automatically activating relevant devices without waiting for a command (e.g., setting the ideal temperature when approaching home), users will be freed from unnecessary micro-management effort. This means reducing cognitive load and maximizing comfort (Singh et al., 2014).

Energy Efficiency and Sustainability: Personalized recommendation systems can learn not only comfort but also energy consumption habits. By offering contextual suggestions that increase user energy efficiency without sacrificing comfort (e.g., "If the blinds are closed at this hour, the air conditioner will run 10% less"), both individual costs will be reduced and a contribution to environmental sustainability will be made (Balta-Ozkan et al., 2014).

Societal and Industrial Importance: The success of this project can increase the rate of industrial adaptation by offering a solution to the usability problem of smart home technologies. The personalization engine to be developed provides a valuable infrastructure that can serve as an enabling technology for many advanced projects such as health monitoring, elderly care, and security scenarios (Alaa et al., 2017).

1.3.2 Personal and Scientific Areas of Interest

This research holds particular scientific interest because it is located at the intersection of the rapid progress made in the fields of the Internet of Things (IoT) and Artificial Intelligence (AI) in recent years.

Technological Maturity: The widespread adoption of IoT devices is coupled with the low latency and high bandwidth brought by 5G mobile communication technology. This indicates that the hardware requirements for processing large datasets in real-time and for the entire smart home system have reached a point of capability.

The Role of Artificial Intelligence: Thanks to evolving Machine Learning (ML) and Deep Learning (DL) techniques, it has become possible to extract complex user behavior patterns from sensor data. This thesis focuses on the mission of having devices now serve people with their full performance, utilizing the synergy created by technology reaching this suitable habitat. This constitutes a pioneering study for future context-aware systems (Schmidt et al., 1999).

1.3.3 Related Work and Literature Review

Scientific efforts towards personalized smart home interaction are generally categorized under Context-Aware Systems and Recommendation Systems.

Context-Awareness Focused Studies: The literature has focused on enabling smart home systems to perceive user context (location, time, activity). Reviews on Context-Aware Tools (Schmidt et al., 1999) have detailed how systems leverage environmental data to improve the quality of user interaction. Machine learning- based approaches have been developed with the aim of predicting user requests in advance and providing the appropriate service (Sobhy, n.d.). These studies have generally emphasized the sensing and prediction phases.

Personalized Recommendation Systems (PRES): Traditional recommendation systems (Collaborative Filtering and Content-Based Filtering) have been adapted for smart home systems. Sobhy's work on recommendation systems addressing multi-user environments aimed to resolve conflicts by taking different user preferences into account. These systems are designed to provide personalized outputs in specific areas such as comfort and energy management (the PDSRS-LD study used DL for sleep patterns). However, in most current studies, the mechanism for the recommendation engines to provide real-time, two-way, and personalized feedback via a mobile application is not fully integrated.

Identification of the Gap: This thesis aims to bridge the gap in the existing literature by combining context-awareness and recommendation mechanisms, and presenting these outputs to the user through a mobile interface in a customized, actionable form.

In this project, by closing the identified gaps in existing systems, a mobile application interface will be created that takes the raw data provided by IoT devices, processes it with artificial intelligence, and generates personalized and proactive recommendations.

This thesis aims to address the shortcomings in proactive personalization and contextual feedback within the existing mobile interaction interfaces of Smart Home systems. The project aims to provide significant scientific contributions at both the theoretical and applied levels by integrating the fields of IoT, Machine Learning (ML), and mobile application development.

1.4 Goal/Contribution

1.4.1 Core Objectives of the Study

The primary goals of this thesis are to develop and validate the following key components to solve the identified problem:

Develop a Personalized Context Sensing Model: The aim is to design a data modeling infrastructure capable of analyzing the instantaneous context (location, time, and activity) and habit data of multi-users in the home.

Create a Behavior Prediction and Recommendation Engine: The objective is to develop a Machine Learning-based engine that uses the developed data model to predict the user's likely future device interactions (e.g., "The brightness of the lights will soon be reduced to 50%") with high accuracy and generates proactive recommendations based on this prediction.

Design and Implement a Personalized Mobile Application Interface: The aim is to design and implement an Android/iOS-based mobile application that presents the outputs of the developed prediction engine to the user in an intuitive and actionable format. This interface will offer not only control but also personalized recommendation streams.

Evaluate System Effectiveness: The goal is to experimentally prove the effectiveness of the developed system—compared to traditional reactive control systems—in terms of energy efficiency, user satisfaction, and prediction accuracy, using empirical methods (conducting tests on a prototype).

1.5 Project Scope

1.5.1. Key Activities within the Scope

Data Modeling and Collection Infrastructure Setup

IoT Integration: Collection of raw usage and contextual data (timestamp, device status, energy consumption, etc.) from commercially available IoT devices operating with Wi-Fi or similar protocols (e.g., smart plug, thermostat).

Multi-User Data Structure: Designing a personalized data labeling and storage mechanism capable of distinguishing the habits of multiple users (Multi-User) in the home.

Contextual Data Collection: Obtaining external contextual data that influences user behavior (e.g., time of day, number of users in the home, weather conditions) by the system.

Habit Prediction and Recommendation Engine Development (ML Model)

Training the Behavior Model: Selecting and training a Machine Learning (ML) algorithm that uses the collected and processed historical data to predict the user's future device interactions (time of use, preferred setting) with high accuracy.

Proactive Recommendation Generation: Creating actionable recommendations (e.g., "It is recommended to lower the thermostat ") that enhance user comfort and/or save energy, based on the outputs obtained from the prediction engine.

Mobile Application Prototype Development

Platform Selection: Developing the mobile application, which will communicate with the recommendation engine in real-time, on Android and iOS platforms.

Personalized User Interface (UI): Designing and implementing an intuitive interface that displays not only reactive control (on/off) but also a personalized proactive recommendation stream.

Two-Way Communication: Enabling the application to both transmit device control commands and receive and present feedback (recommendations) from the recommendation engine to the user.

Performance Evaluation

Experimental Validation: Evaluating the developed prediction algorithm in terms of accuracy and precision metrics.

System Comparison: Comparing the new proactive system with traditional static/reactive smart home systems in terms of user satisfaction and potential energy efficiency.

1.5.2. Parts to be Excluded from the Project (Out-of-Scope Areas)

Special Hardware Development: Developing custom IoT hardware from scratch is outside the scope of the thesis; existing commercial or open-source devices will be used.

Extensive Field Tests: Large-scale and long-term field tests for the widespread deployment of the application will not be conducted; results will be validated with a limited user group or simulated data sets.

Emotion and Mood Analysis: The recommendation mechanism will be based only on repetitive time and context-based habit data. Providing recommendations by analyzing the user's current emotional state or mood is excluded from the scope of this study and will not be performed.

Web Interface Development: The study will focus only on the mobile application prototype; a web-based control panel will not be developed.

1.5.3 Project Methodology

Device Selection and Protocol: In our project, we plan to use a smart lamp that already has a smart application and which we will not struggle to connect to. The device must support an API (Application Programming Interface) or the MQTT (Message Queuing Telemetry Transport) protocol.

Data Collection: When the device (lamp) is turned on or off, its brightness changes, or it consumes energy, it sends this data over the internet to a Central Server (Cloud Server).

Recommendation Engine: The Machine Learning model we develop will run on this central server. It analyzes the collected habit data and predicts the next action (e.g., brightness will drop to 30% at 20:15).

Flutter Application Communication:

Control: When the Flutter application (the user's mobile phone) wants to control a device (e.g., "Turn Off"), it sends this request directly to the Cloud Server. The server then sends the command to the relevant device (the lamp).

Receiving Recommendations: When the recommendation engine makes a prediction, it transmits this prediction (e.g., "You should lower the lamp setting to 30%") to the Flutter application as a real-time notification, again via the server. The Flutter application displays this notification to the user ("Suggestion: Should we lower the brightness?").

1.5.4 Broad Application Area

Although the application area of this thesis is primarily defined as the residential environments focused on by Smart Home systems, the personalized habit prediction and recommendation engine principles developed can be extended to all types of small-scale automation areas where Internet of Things (IoT) devices are used. Theoretically, this study provides an adaptable infrastructure for personalized energy management and comfort optimization scenarios in smart offices, limited laboratory environments, hotel rooms, or other enclosed spaces. However, the experimental validation and prototype development stage will be carried out on a limited residence/home simulation due to the need for modeling multi-user habits.

1.6 Standards, Ethics, Constraints And Conditions

1.6.1 Standards and Ethics

Standards and Protocols Used: This study will reference open standards widely used in the field of the Internet of Things (IoT). Specifically, the MQTT (Message Queuing Telemetry Transport) protocol will be used for inter-device data transfer and real-time messaging due to its lightweight and flexible structure. Software engineering guidelines from the IEEE and relevant design patterns will be the basis for the software development process.

Ethical Rules and Personal Data Protection: Since the application will process sensitive personal data regarding users' in-home behavior (habit data, device usage times), full compliance with the Personal Data Protection Law (KVKK/GDPR equivalent) will be ensured during the processes of data collection, storage, and processing. All data sets will be anonymized and used only for research purposes. Obtaining informed consent from users before data collection is paramount among the ethical guidelines.

1.6.2 Constraints and Conditions

Technical Constraints (Hardware, Software, and Time): The project faces hardware limitations such as the limited processing power and network latency of commercially available IoT devices used for prototyping purposes. Furthermore, due to the time constraint imposed by the thesis duration, the amount of real user data used to train the recommendation engine will be limited, and thus a large part of the model will be supported by simulated data sets. The Flutter framework chosen for the mobile application provides cross-platform compatibility but may occasionally create limitations in accessing native device functions.

Other External Conditions: The experimental validation of the study will be conducted on a limited residence/home simulation. This means that the results are valid in a controlled laboratory environment, but the potential for scaling to a wide geographical distribution or hundreds of users is outside the scope of this thesis.

CHAPTER TWO

LITERATURE REVIEW

2.1 Related Works

2.1.1 Intelligent Temperature Control System Utilizing Gated Recurrent Units and the Internet of Things

Studies utilizing advanced machine learning to enhance the potential for proactive control in smart environments have been included in the recent literature. For instance, an intelligent temperature control system has been successfully demonstrated by Lu and Shin (2024) through the integration of IoT and GRU. Their methodology has been crucial in establishing that the GRU model can be effectively utilized for predicting user behavior for a specific appliance, which has resulted in reduced energy consumption and enhanced comfort. However, the work by Lu and Shin (2024) has been restricted to a single control domain (temperature) and has been found to lack the generalized system architecture required for comprehensive multi-device and multi-user personalization across the entire home environment. Critically, the challenges related to multi-user habit separation and the development of a unified mobile interface for actionable recommendations have not been addressed in their implementation. Consequently, the need for a holistic, mobile-centric solution for proactive, system-wide interaction has been established as the research gap for the present thesis.

2.1.2 Unsupervised Segmentation of Smart Home Logs for Human Habit Discovery

An unsupervised methodology has been proposed to automatically segment raw sensor logs into Human Habits, aiming to eliminate the need for manual labeling in smart home systems. In this approach, a discretization strategy applied to the timestamp attribute of

the sensor logs has been guided using quality measures of Petri nets, such as simplicity and structuredness. The methodology has provided visual process models (Petri nets) of habits and has been claimed to be the first study in the field to achieve automatic segmentation of a full smart space log. However, this work has only been evaluated against a single-human dataset. The problem of attributing each measurement to the corresponding resident in multi-human environments has limited the practical applicability of the solution. Furthermore, the extracted models have been used for offline analysis and recognition purposes, not for prediction; this has indicated that the study does not directly serve the thesis's goal of proactive and real-time mobile recommendations.

2.1.3 Secured Smart Mobile App for Smart Home Environment

A new method has been proposed to solve the security problems arising from the single-stage authentication processes of existing smart home appliances. The proposed SSMASHE (Secured Smart Mobile App for Smart Home Environment) system has provided a two-stage authentication procedure to ensure the security of the devices. This system has been constructed from three main modules: a smart mobile app, smart devices, and a big data platform. The effectiveness of the method has been measured through the delay time in the authentication process, and it has been proven to be less than existing methods like FIDO. However, this study has been focused entirely on security, authorization, and delay optimization. Objectives such as user habit learning, behavior prediction, or proactive mobile recommendation using device usage data have not been addressed in this system.

2.2 Comparison with the Existing Solutions

2.2.1 Mobility Predictions for IoT Devices Using Gated Recurrent Unit Network

A system that is a hybrid of Principal Component Analysis (PCA) and Gated Recurrent Unit (GRU) algorithms has been proposed to enhance the mobility (tracking and

trajectory prediction) of IoT devices in wireless urban areas. In this approach, PCA has been primarily applied to reduce signal noise and the number of features in high-dimensional data sets, and this step has optimized the overall system performance. Within the literature, it has been stated that deep learning techniques, unlike shallow learning algorithms, offer higher learning capacity for complex and dynamic data. In the experimental evaluations conducted, it has been proven that the PCA-GRU model has exhibited superior performance over PCA-MLP and PCA-LSTM models in terms of tracking and trajectory predictions. This study has justified the decision that the GRU algorithm is the most technically suitable option for the thesis's complex IoT habit data and time-series predictions.

2.2.2 Human Activity Recognition From Smart Watch Sensor Data Using a Hybrid of Principal Component Analysis and Random Forest Algorithm

The detection and classification of daily human movements using smart watch sensor data has been aimed; eight different daily activities have been examined within this scope. Principal Component Analysis (PCA) has been applied to the dataset in order to reduce feature dimensionality and increase classification accuracy. In this developed hybrid approach, methods including Random Forest (RF), Support Vector Machine (SVM), C4.5, and k-Nearest Neighbor (kNN) have been compared as the classification algorithms. As a result of the experimental evaluations conducted, it has been proved that the most successful outcome in classification performance has been obtained from the Random Forest method. This success has demonstrated that Random Forest is a superior alternative to traditional shallow learning algorithms for the classification and decision-making stages of habit data.

2.2.3 Novel Bayesian Inference Algorithms for Multiuser Detection in M2M Communications

The problem of Multiuser Detection (MUD) has been stated to be a challenging task in supporting massive Machine-to-Machine (M2M) communications, primarily due to the

user activity factor being typically unknown in practical applications. To overcome these difficulties, novel Bayesian inference algorithms that do not require knowledge of the activity factor have been proposed. Among the proposed methods, the Gaussian Mixture Model (GMM) method has been shown to exhibit better performance than all existing schemes in both low and high activity factors by effectively utilizing prior knowledge, such as zero transmitted signals and finite-alphabet constraints. This outcome has supported the conclusion that Bayesian inference-based approaches offer more reliable results than traditional methods, despite the uncertainty regarding the number and activity status of users in complex multi-user environments.

2.2.4 Comparison of Flutter and React Native Platforms

The study has aimed to provide a comparative analysis of Flutter and React Native, which are cross-platform tools offering solutions to the time loss and high cost issues inherent in mobile application development processes. The two platforms have been compared under six different technical headings, including their architectures, utilized languages (Dart/Javascript), popularity, and community support (GitHub, Stackoverflow data). Although both platforms have been determined to possess unique advantages, Flutter has been found to be superior in certain aspects: The application size has been shown to be smaller (4.7 MB versus 7 MB), the application's initial launch speed is higher, and Flutter has been ranked first among cross-platform structures in terms of developer interest (75.4% approval rate, surpassing React Native's 62.5%). This work, therefore, provides a strong basis proving that the selection of Flutter as the mobile interface for the thesis project is the correct technical choice in terms of meeting the requirements for low application size and high-performance in a proactive smart home system.

2.2.5 Analysis of Smart Home Systems in the Context of the Internet of Things in Terms of Consumer Experience

The Technology Acceptance Model (TAM) and a customer experience scale have been integrated to improve models for consumers' acceptance of smart home technologies. Research results have shown that psychological factors, such as perceived intelligence and perceived ease of use, have a significant effect on the intention to use IoT systems via perceived usefulness. It has been emphasized that, to increase the perceived usefulness gained by consumers from the system, manufacturers must develop a highly intelligent system that automatically sets the best alternative. This situation has demonstrated that manufacturers need to design smart IoT systems with self-learning capabilities. Therefore, this research has proven that the thesis's goals of providing high-accuracy prediction (GRU) and proactive recommendations are critical to delivering the Perceived Intelligence factor expected by consumers from the system.

2.2.6 A Review of IoT Applications in Healthcare

The integration of Internet of Things (IoT) technologies in the healthcare industry has been highlighted as a transformative shift with tangible benefits. Within this context, the role of Artificial Intelligence (AI) and Machine Learning (ML) algorithms has been underscored for their capability to analyze vast amounts of IoT data, thereby enabling the prediction of disease progression and the suggestion of individualized treatment strategies. Furthermore, the necessity of utilizing modern infrastructure like Edge Computing and 5G networks has been established to address challenges related to network latency and to facilitate real-time data analysis crucial for timely interventions. This research has demonstrated that technological advancements in IoT support proactive intervention and personalized care, which justifies the current study's focus on developing an advanced, prediction-based system over conventional, reactive smart home systems.

2.2.7 A Systematic Comparison Between Flutter and React Native from Automation Testing Perspective

The main goal has been to systematically compare the reusability, integration, and compatibility of automation testing across cross-platforms. Identical "To-Do List"

applications have been created in both Flutter and React Native, and subsequently tested on Android devices using TestProject.io. React Native has outperformed Flutter in the categories of test reusability and compatibility. However, integration has been found to be 100% successful on both platforms. This work adds a constraint to the literature, suggesting that structures generated in Flutter may face difficulties in having their UI elements correctly detected by automation tools.

2.2.8 Monitoring Home Security by Developing an IoT-Based Smart Home Automation System Controlled by Android App and Web

The aim of the study has been defined as designing a cost-effective IoT-based smart home automation system by utilizing Raspberry Pi 4, Flutter, and Firebase to enhance security, energy control, and comfort. Raspberry Pi 4 has been utilized as the central controller, Firebase for real-time synchronization, and Flutter has been used for the mobile/web interface. Fingerprint authentication and gas leak detection have been established as core functionalities. The system has demonstrated reliable performance across all features with zero false positives/negatives during testing. The versatility of Flutter has been shown to enable an attractive and responsive user interface across both mobile and web platforms using a single codebase.

2.2.9 Development of Water Level Monitoring Application in Smart Home Systmes Using Flutter

The main goal has been defined as developing a real-time water level monitoring system utilizing ESP8266 + HC-SR04 + MQTT + Flutter. Sensor data (water level) has been collected using the ESP8266, transmitted to the mobile application via MQTT, and subsequently visualized across six different platforms (Android, iOS, Web, etc.) using Flutter. The system has been proven to provide accurate and real-time water level information, and the application's ability to operate flexibly across six different platforms

has been demonstrated. This work validates the high cross-platform flexibility and real-time data transmission reliability of the project's IoT-MQTT-Flutter technology stack.

2.2.10 Developing a Multiprofessional Mobile App to Enhance Health Habits in Older Adults: User-Centered Approach

The aim of the study has been to develop a multiprofessional mobile application focused on promoting healthy habits in older adults, adhering to the principles of software engineering and a User-Centered Design (UCD) approach. The development and refinement process has been conducted using a mixed methods approach, actively involving participants through three phases of focus groups and design thinking. After adjustments based on participant feedback, satisfaction scores (SAM) have been increased from 34.89% to 70.65%, and usability scores (SUS) have significantly risen from 71.23 to 87.14, demonstrating excellent usability. Participant feedback has emphasized the need for large font sizes, simplified navigation, visual confirmation signals, and the ability to integrate with external health devices. This work, therefore, strongly supports the thesis goal of employing a user-centered approach to enhance the usability and personalized integration of the Smart Home interface.

2.2.11 Who's In Control?: Interactions In Multi-User Smart Homes

The study has aimed to systematically understand the interactions, cooperation, and tensions that arise among multiple people and devices in contemporary, deployed smart homes. A mixed-methods study has been conducted, combining semi-structured interviews and experience sampling with 18 participants. The findings have revealed a significant power imbalance, primarily emphasizing the outsized role of the person who installs devices ("Smart Home Driver") in terms of selection, control, and data access compared to passive users. Furthermore, tensions have been observed between parents and children regarding control, and many users still prefer analog controls. This work, therefore, critically supports the thesis's core objective of addressing the existing literature gap by developing a personalized, multi-user system that minimizes these power

differences and addresses the limited multi-user support currently provided by commercial smart home platforms.

2.2.12 Better to Ask Than Assume: Proactive Voice Assistants' Communication Strategies That Respect User Agency in a Smart Home Environment

The aim of the study has been to investigate the verbal communication strategies that Proactive Voice Assistants (VAs) should employ to respect user agency while taking predictive action in a smart home environment. The research has been conducted using the Wizard of Oz (WoZ) method with 20 participants to simulate a proactive smart home. The study has concluded that proactive systems should not make blind assumptions; instead, they must explicitly ask for confirmation or provide justification for the proactive action to maintain user control and foster trust. This work, therefore, critically supports the thesis's design goal for the Flutter mobile interface, affirming that the ML prediction output must be presented as a request for approval to enhance user agency, rather than being automated directly.

2.2.13 Real-Time Energy Management in Smart Homes Through Deep Reinforcement Learning

A real-time, model-free Home Energy Management (HEM) approach, grounded in the Proximal Policy Optimization (PPO) algorithm, has been presented to optimize load scheduling and manage diverse loads, including Electric Vehicles (EV) and Energy Storage (ES), under stochastic conditions. The optimization problem has been modeled as a Markov Decision Process (MDP). The proposed methodology has been validated using real-world datasets, demonstrating a substantial 31.5% reduction in electricity bills. Furthermore, the algorithm has been shown to proficiently control bidirectional converters, enhancing the overall power factor of the smart home from 0.44 (benchmark) to 0.901. This work, which marks the inaugural application of a real-time, model-free technique for optimal active and reactive power management in HEM contexts, supports

the thesis's use of advanced ML for achieving superior economic and technical efficiency in smart home prediction systems.

2.2.13 Deep Reinforcement Learning for Real-Time Energy Management in Smart Home

A real-time Home Energy Management (HEMS) algorithm, based on the Proximal Policy Optimization (PPO) approach, has been proposed to minimize energy costs while ensuring user comfort amidst uncertainties such as weather and real-time electricity prices. The HEMS problem has been formulated as a Markov Decision Process (MDP). A novel Mixed Output Policy (MOP) has been designed within the PPO agent to handle both discrete (shiftable appliances) and continuous (controllable appliances) control actions simultaneously. Simulations based on real-world data have demonstrated that the proposed method reduces cumulative energy costs by 42.9% compared to the baseline and exhibits superior robustness against thermal disturbances when compared to other DRL algorithms like DDPG and TRPO. This work strongly validates the thesis's use of PPO for real-time operation and supports the necessity of designing a prediction engine capable of handling mixed control actions in the smart home environment.

2.2.14 Comparison with HTTP and MQTT In Internet of Things (IoT)

The aim of the study has been to compare the lightweight MQTT and the traditional HTTP protocols based on speed, complexity, and power consumption to determine the best communication protocol for IoT devices. The protocols' architectures have been analyzed, contrasting MQTT's Publish/Subscribe model with HTTP's Request/Response model. The results have shown that MQTT is superior in various aspects: it has consumed less power to maintain an open connection, has provided higher throughput (90 times faster than HTTP in 3G networks), and has achieved a high message delivery rate. This work strongly validates the thesis's decision to use the MQTT protocol as it addresses the

essential IoT requirements of low latency and minimal bandwidth for real-time data exchange in the smart home system.

2.2.15 Smart Home Management Based on Deep Learning: Optimizing Device Prediction and User Interface Interaction

The aim of this work has been to address the challenges in smart home systems, including the accuracy of device status prediction and user interface (UI) design. An innovative multimodal prediction model combining Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) networks has been proposed to enhance predictive accuracy. This predictive model, alongside the Deep Q Network (DQN) for intelligent control, has been deployed utilizing Cloud Computing for an intuitive UI and rapid response. Experimental results have demonstrated that the proposed system significantly reduces prediction errors (MSE by 20%, MAE by 15%). Furthermore, the system has achieved a low response latency of 85 ms and increased user satisfaction by 25% compared to traditional models. This study validates the thesis's core approach, confirming the superiority of a CNN-LSTM hybrid model for multimodal data processing and reinforcing the critical link between efficient cloud-backed AI and high UI satisfaction.

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CHAPTER THREE

This section outlines the necessary specifications, features, and constraints for the successful implementation and functionality of the Personalized Smart Home System. These requirements provide a clear foundation for system design, development, and evaluation, ensuring alignment with project objectives and stakeholder expectations. The requirements are categorized into functional and non-functional requirements, addressing both the core operations of the system (User management, IoT control, AI recommendations) and the quality attributes essential for its efficiency, usability, and reliability.

The system operates on a mobile platform integrated with a cloud backend and a local gateway (Home Assistant), ensuring seamless interaction between residents, administrators, and IoT devices.

3.1. User Roles and Actors

The system defines two primary roles with shared capabilities for device control but distinct administrative privileges.

3.1.1. Resident (Standard User)

Residents are users who have joined a home environment. They have full authority to control devices and receive recommendations but lack administrative privileges regarding the home's existence.

Device Control: Can operate all devices (turn on/off, adjust settings) within the home.

Device Management: Can rename devices for better personalization.

Recommendation: Receives personalized suggestions via the mobile interface.

3.1.2. Home Administrator (Admin)

The user who creates the home environment is designated as the Admin. In addition to all Resident privileges, the Admin has exclusive rights to:

Home Management: The ability to permanently delete the home environment from the system.

Member Management: Authority to approve join requests or remove members from the home.

3.2. Functional Requirements

3.2.1. Registration, Login, and Home Management

Unique Identification: The system must generate a unique 8-character ID for each user (starting with 'P') and each home (starting with 'H') to ensure distinct identification in the database.

Join Requests: Users must be able to send a request to join a home using its unique ID and password. Access is granted only upon Administrator approval.

Device Renaming: All authorized users (Residents and Admins) must be able to rename devices within the application interface to facilitate easier recognition.

3.2.2. IoT Device Control and Integration

Home Assistant Integration: The system must communicate with IoT devices exclusively through the **Home Assistant API**. The mobile application shall not communicate directly with end devices but will route commands through this standardized gateway.

State Synchronization: The application must display the real-time status of devices (e.g., "On/Off", "24°C"). If the application is running in the background, it must show the last known state upon reopening.

Inactivity Auto-Shutdown: To ensure energy efficiency and safety, the system must monitor user interaction. If no interaction (control command) is detected within the application for **12 consecutive hours**, the system shall automatically switch to "Passive Mode" and turn off all connected active devices.

3.2.3. AI and Recommendation Engine

Cold Start Processing: The recommendation engine must be capable of processing a simulated dataset of at least **10,000 records** to generate initial "average human" recommendations for new users based on their BMI, age, and location.

Proactive Notifications: Recommendations must be delivered as **Push Notifications** when the application is closed or running in the background.

In-App Recommendation Tab: A dedicated section within the application must list active and past recommendations for user review.

3.3. Non-Functional Requirements

3.3.1. Performance

Control Latency: The latency between a user issuing a command (e.g., "Turn on light") on the mobile app and the physical device executing the action must be less than **3 seconds** under normal network conditions.

Recommendation Speed: The AI engine must analyze context changes and generate a personalized recommendation within **20 seconds** of a trigger event (e.g., a change in weather or user activity).

3.3.2. Availability and Failure Handling

User Internet Disconnection: If the user's mobile device loses internet connectivity, the IoT devices at home must continue to operate in their **last known state** (they should not turn off automatically).

Device Internet Disconnection: If the IoT devices/home gateway lose internet connectivity, the system must trigger a **Fail-Safe** mechanism where the devices automatically turn **OFF** to prevent unsupervised operation.

Application Offline State:

If the app is running in the background and loses internet, it must display the last known state of devices.

If the app is cleared from memory (cache cleared) and restarted without an internet connection, it must display a "No Connection" error and prevent access to control interfaces.

3.3.3. Scalability

Data Handling: The backend architecture must be scalable to support the processing of the initial 10,000-entry dataset for the cold-start algorithm without system degradation.

3.3.4. Security

Role-Based Access Control: Critical actions such as "Deleting a Home" must be strictly restricted to the User ID associated with the Home Administrator role.

API Security: All communications between the Mobile App, Cloud Server, and Home Assistant Gateway must be encrypted to prevent unauthorized interception of commands.

3.4. Data Acquisition and Preprocessing Requirements

This section defines the strict data formatting and cleaning rules required to prepare raw IoT logs for the pattern recognition engine.

3.4.1. Time Resolution and Normalization

Minimum Time Unit: The system shall not process or store data with second-level precision. All incoming timestamps must be rounded to the nearest **minute**.

Duration Formatting: To maintain consistency in user reporting and data storage, durations shall not be stored solely as total minutes (e.g., 75 min). Instead, the system must convert and store these values in an "**Hours and Minutes**" format (e.g., 1 Hour 15 Minutes).

Rounding Logic: Any event lasting less than 60 seconds shall be rounded up to 1 minute to ensure minimal usage is captured.

3.4.2. Contextual Data Segmentation

The system must tag every logged event with temporal context attributes to enable multi-layer pattern recognition:

Daily Context: Each log must be associated with a specific date and time slot.

Weekly Context: Each log must be tagged with the specific day of the week (Monday, Tuesday, etc.) to identify weekly cyclic habits.

Monthly Context: Each log must be tagged with the day of the month (e.g., "2nd day of the month") to identify monthly recurring behaviors.

3.5. Recommendation Engine Specifications

This section details the core logic of the Habit Detection Algorithm, specifically how it identifies patterns across different time horizons and manages the lifecycle of a "Habit."

3.5.1. Multi-Layer Pattern Recognition

The engine must scan for patterns across three distinct temporal layers:

Daily Routine Scan: The system shall analyze the data at the end of each day to identify actions repeated consistently at the same time (e.g., "Every day at 07:00").

Weekly Cycle Scan: The system shall check for patterns that occur only on specific days of the week (e.g., "Every Tuesday at 19:00").

Monthly Recurrence Scan: The system shall analyze the data to detect actions repeated on specific dates of the month (e.g., "Every month on the 2nd day").

3.5.2. Habit Acceptance and Rejection Thresholds (Hysteresis Logic)

To prevent system instability and spam notifications, the system shall implement a **Dual-Threshold Mechanism** for classifying a behavior as a "Habit."

Habit Formation Threshold (Entry > 60%): A repeated action shall only be classified as a "Habit" and trigger recommendations if the probability of occurrence exceeds **60%**.

Example: If the user turns on the heater at 20:00 for 4 out of the last 6 observed days (~66%), it becomes a Habit.

Habit Termination Threshold (Exit < 45%): Once a behavior is classified as a Habit, it shall remain active until its probability score drops below **45%**.

Rationale: This hysteresis gap (15%) prevents a habit from being constantly added and removed due to minor deviations in user behavior.

3.5.3. Recommendation Triggering

Prediction Match: The system shall compare the current context (Time, Day, Date) with the active "Habit List."

Actionable Output: If a match is found with a probability > 60% (or an existing habit > 45%), the system must generate a notification proposing the likely action (e.g., "It's Tuesday evening; would you like to set the living room mode to 'Cinema'?").

3.6. Model Lifecycle and Data Retention

This section defines how the system manages storage limits and updates its knowledge base over time.

3.6.1. Update Frequency

Daily Processing: The model must process new logs and update habit probabilities at the end of every 24-hour cycle.

Periodic Aggregation: The system shall perform a broader scan for Weekly and Monthly patterns at the end of each respective period.

3.6.2. Data Retention Policy (Sliding Window)

To balance performance with historical accuracy, the system shall implement a **6-Month Rolling Window** (Sliding Window) mechanism for data retention.

Capacity: The database shall maintain active detailed logs for exactly **6 months**.

FIFO Mechanism: The system shall operate on a "First-In-First-Out" basis. When the data retention period exceeds 6 months, the logs from the **first (oldest) day** in the dataset must be automatically deleted to make room for the **new (current) day**.

Model Continuity: The deletion of the oldest day shall trigger a minor recalculation of habit probabilities, ensuring the model always reflects the most recent 6-month trend without requiring a full reset.

CHAPTER FOUR

Design

4.1. Architectural View

4.1.1 Scenarios

Personalized Recommendation Flow : Based on past logs where User A said "Read a book" in the mobile app, the system proposes to automatically set the lamp's brightness to a suitable reading level at 9:55 PM.

Multi-User Device Control: User A and User B at home control the same shared device (e.g., Thermostat), and the changes made by one user are instantly updated in the other user's mobile app.

New Device Addition and Recognition: The user adds a new smart lamp to the app, and the device successfully connects to the IoT Platform, allowing the system to recognize it as a command target.

4.1.2 Logical View

User Class: userID (PK), fullName, email, height, weight, environmentID (FK).

Environment Class: environmentID (PK), address, creationDate. (Groups all users and devices in a home.)

Device Class: deviceID (PK), environmentID (FK), type (Lamp, Thermostat), status (On/Off), lastModificationDate.

Habit Class: habitID (PK), userID (FK), habitType (Drink Water, Reading), targetValue, currentValue, recommendationFrequency.

BehaviorLog Class: logID (PK), userID (FK), deviceID (FK), action (TurnOn, TurnOff), timestamp, parameters (Brightness: 70%).

4.1.3 Process View

Service	Concurrency Type	Inter-Process Communication (IPC)	Role in Data Flow
Flutter	Multi-User	REST API (via API Gateway)	Command Launcher
API Gateway	Concurrent	REST API (Synchronous),	Router
IoT Control Service	Asynchronous	MQTT (with devices), REST API (with API Gateway)	Device Status Manager
Recommendation Engine Service	Asynchronous	Kafka (Log Reading), REST API (Result Publishing)	Critical Processor (CPU Intensive)

4.1.4 Development View

Component	Technologies / Languages	Repository	Layer Relationship
Mobile Application	Flutter, Dart	Repo A: smart_home_flutter_app	Presentation Layer
API Gateway	Python (FastAPI)	Repo B: api_gateway_service	Business Logic Layer
Recommendation Engine Service	Python (Scikit-learn/Pandas)	Repo C: recommendation_engine	Business Logic Layer (ML/Data Processing)
IoT Control Service	Python (Paho-MQTT/AsyncIO)	Repo D: iot_control_service	Business Logic Layer (Real-Time Communication)
Shared Libraries	Python / Dart	Repo E: shared_libraries	All Services

4.1.5 Physical View

Component Type	New Deployment Environment	Why It Is Better
IoT Control Service	Kubernetes/Docker (in the Cloud)	Moves the translation work to the Cloud. Processing data from Vestel (e.g.) and sending the command is now handled entirely by your services in the Cloud.
IoT Platform	Google Cloud IoT / AWS IoT	Used only for your own logs and event streams.
Device Connection	External APIs (Google Home/Alexa/Manufacturer API)Cloud to Cloud(C2C)	Your system communicates not directly with the devices, but via an API in the Cloud.

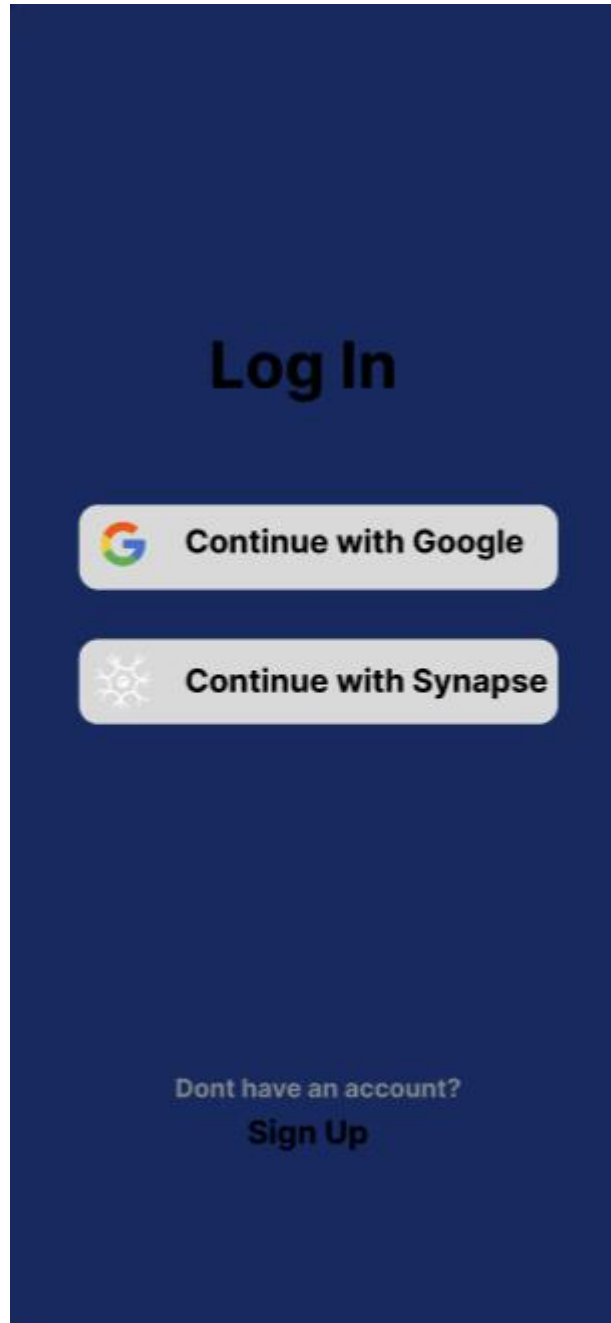
4.2 UI Design



4.2.1 First Page

The entry point of the application reflects the core philosophy of the system. The name "Synapse" (Figure 4.2.1) is inspired by the junction points that allow neurons in the human brain to communicate with one another. Just as the central nervous system coordinates bodily functions, the Synapse app aims to optimize a user's daily life and

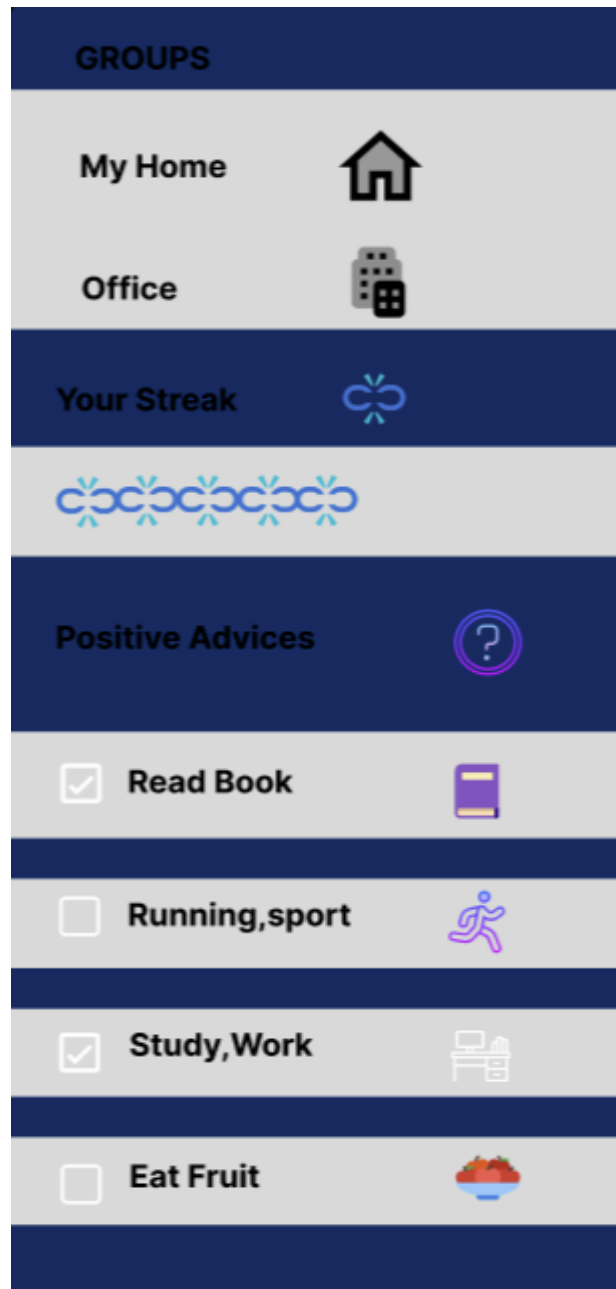
home automation from a central hub. The slogan, "Welcome to a personalized living space experience," reinforces the system's focus on deep personalization.



4.2.2 Log in Page

Designed to ensure secure access, this page offers modern authentication methods(Figure 4.2.2). Users can create a dedicated account within the Synapse ecosystem or utilize third-party services like Google via Single Sign-On (SSO) for rapid registration. During the

initial onboarding, demographic and physical data (such as height, weight, and activity levels) are collected and processed into the database to enhance the accuracy of the recommendation engine during its "Cold Start" phase.

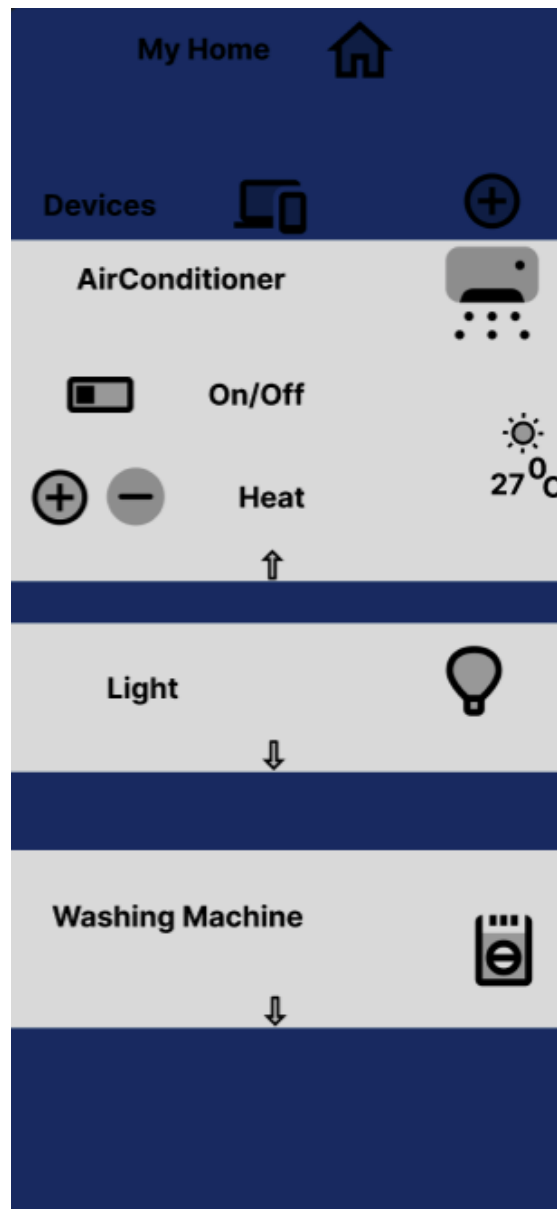


4.2.3 Main Page

Group Management: Users can create specific environment groups for home, office, or other living areas, inviting family members or colleagues to join these shared spaces.

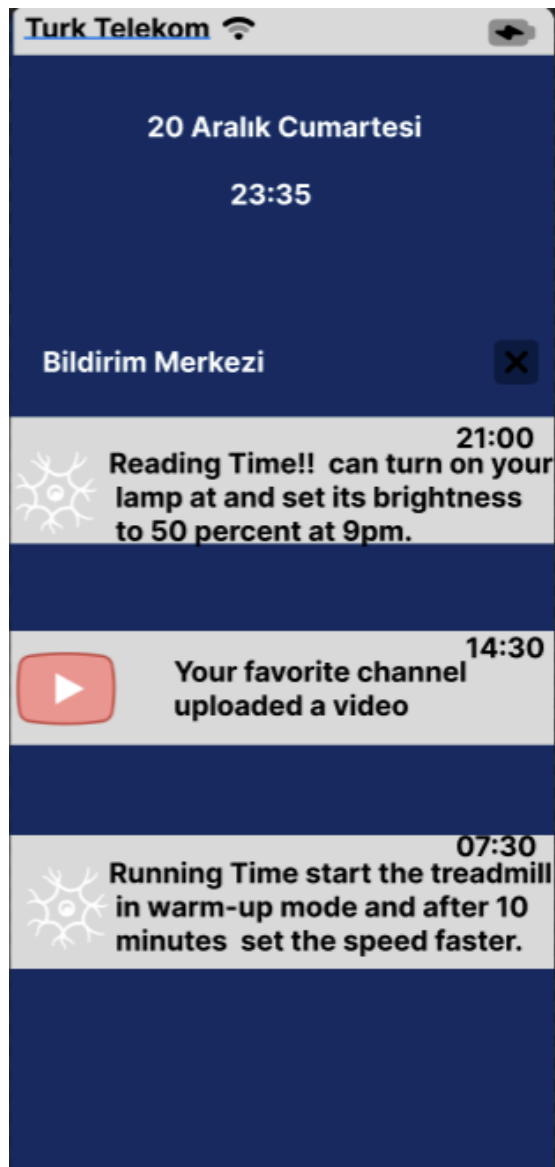
The Streak Mechanism: Gamification techniques are employed to boost user motivation; (Figure 4.2.3) as users complete tasks, they build a continuous "streak."

Positive Advice: The positive suggestions such as reading, fruit consumption, and exercise based on health and productivity literature. As users complete these tasks, the interaction data is analyzed by the Recommendation Engine to learn and adapt to their routines.



4.2.4 Devices Page

This module enables real-time control of integrated smart devices. Through a unified interface, users can manage appliances (Figure 4.2.4) such as air conditioners, lighting, and white goods (e.g., washing machines). Beyond basic on/off toggles, the interface allows users to expand device cards to adjust detailed parameters such as temperature settings, operating modes, or brightness levels facilitated through Cloud-to-Cloud API integration.



4.2.5 Advice Page

The true "intelligent" layer of the system is embodied here (Figure 4.2.5). By analyzing device usage logs and user routines through pattern recognition, the system pushes

contextual suggestions to the phone's notification center. For example, if a user habitually reads at 9:00 PM, the system sends a "Reading Time" notification, proposing to set the lights to reading mode (50% brightness). These are actionable notifications: with a single tap, the user can confirm the suggestion and automatically trigger physical devices to operate with predefined parameters.

4.3 Activity Diagram



4.3.1 Basic Activity Diagram of the Project

Onboarding and Registration The process begins when the user launches the application. The system immediately checks for an existing account through a decision point; if the

user does not have one, they are directed to enter personal metrics such as height, weight, and activity levels to create a profile (Figure 4.3.1). Otherwise, existing users simply log in to proceed.

Navigation to Home Page Once the authentication or registration is successfully completed, the system automatically redirects the user to the main dashboard. This serves as the central hub where the user can access all features of the smart ecosystem.

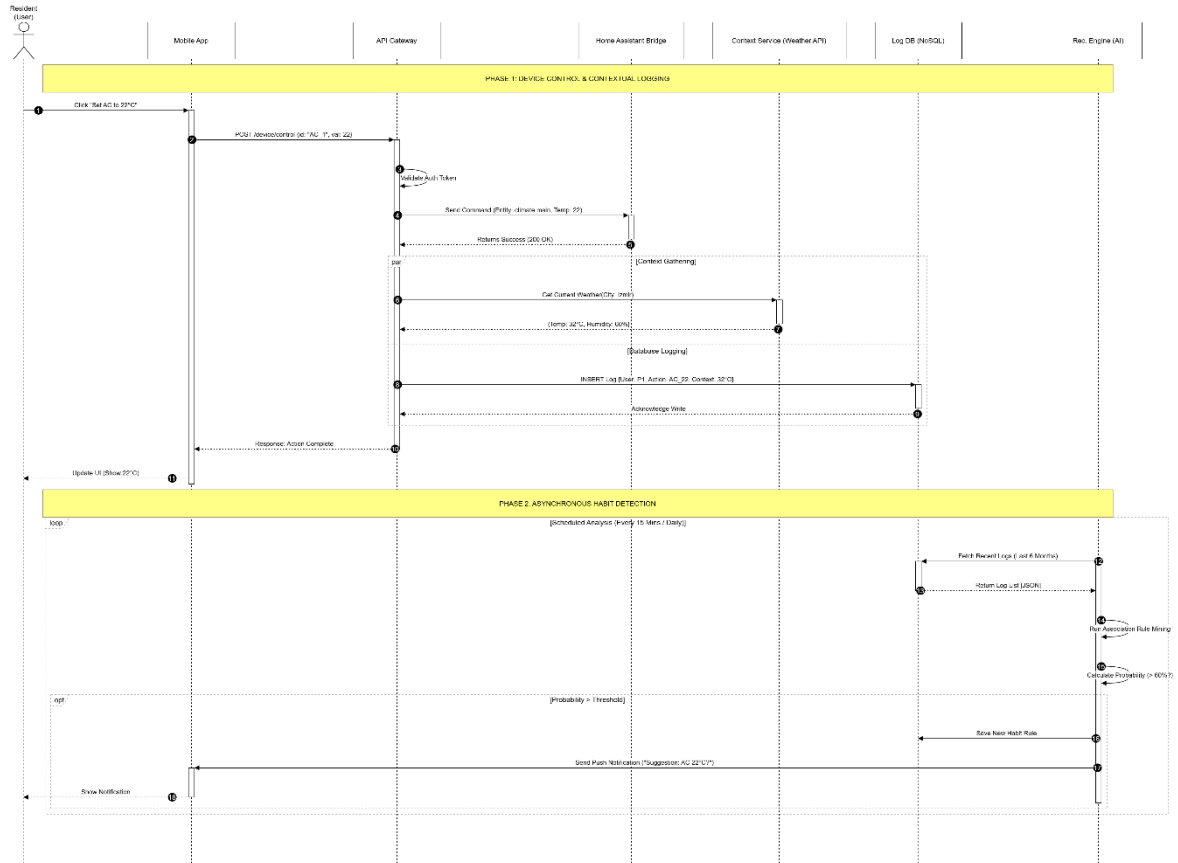
Daily Usage Cycle and Parallel Activities The application supports a parallel workflow, allowing users to engage in multiple activities simultaneously through a "fork" mechanism. On one hand, users can browse positive lifestyle recommendations, complete tasks, and build streaks; on the other hand, they can manage social groups for their home or workplace and exercise direct control over smart devices, such as toggling lights or adjusting the climate. These paths eventually converge back into a unified user experience.

Smart Learning and Recommendation Mechanism In the background, the system constantly analyzes user movements to facilitate machine learning. Even if a distinct habit hasn't been identified yet, the system provides "baseline" suggestions based on the user's initial registration data—such as setting default light brightness to 70%. Once a specific habit is detected, like a user regularly reading at 9 PM, the engine generates a personalized prompt asking if they would like to adjust the environment to their preferred 50% light level.

User Approval and Command Execution Every personalized recommendation is subject to user approval via a decision node. If the user approves the suggestion, the system instantly transmits the command to the relevant device, such as adjusting the lamp. If the user declines, the system records this feedback as a negative sample to refine its learning algorithm and avoid similar mistakes in the future.

Termination and Continuity The process concludes either by maintaining its background operations to ensure continuous monitoring or by terminating when the user closes the application. The system remains in a state of readiness, waiting for the next trigger or user interaction.

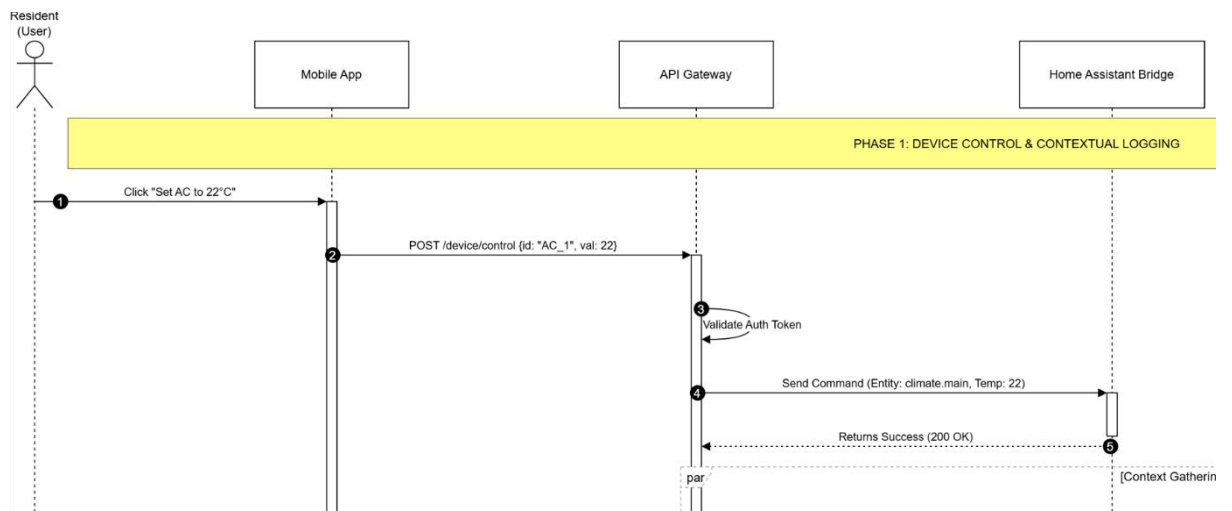
4.4 Sequence Diagram



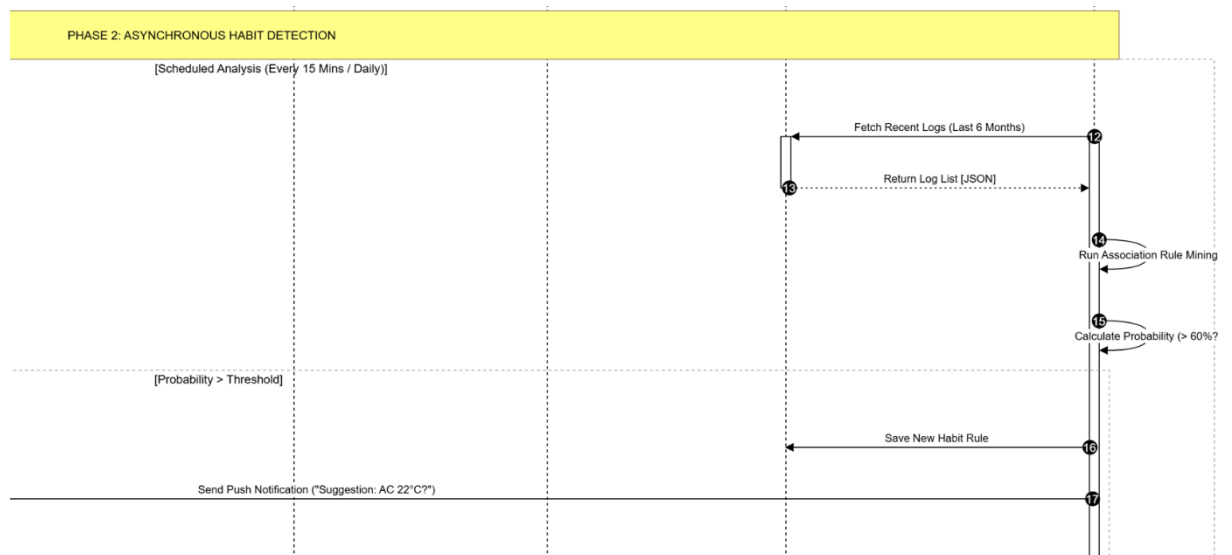
4.4.1 Basic Sequence Diagram of the Project

The sequence diagram presented in Figure 4.4.1 illustrates the end-to-end integration of user-initiated device control operations with the background AI-based learning loop. The process (Figure 4.4.2) initiates when a user modifies a device setting (e.g., an air conditioner). While the API Gateway validates the request and routes it to the physical device via the Home Assistant Bridge, it employs a parallel execution strategy to maintain system performance. This ensures that command execution occurs simultaneously with the retrieval of contextual data (such as external temperature), which is then enriched with the action data and logged into the NoSQL database.

In the second phase of the diagram (Figure 4.4.3), the asynchronous Recommendation Engine is activated. Without impacting the user experience, the system analyzes data from the last 6 months in the background and executes the Association Rule Mining algorithm. If a user behavior repeated within a specific context exceeds the 60% probability threshold, it is registered in the system as a "habit," and the cycle concludes by sending a proactive recommendation (Push Notification) to the user via the mobile application.



4.4.2 Device Control



4.4.3 Asynchronous Habit Detection